TEXORA · LECTURE SHEET

Isomerism

Same formula, different substances: the classification of isomers — structural (chain, position, functional group, metamerism, tautomerism) and stereo (geometrical and optical) — chirality, enantiomers, diastereomers, meso forms, and how to count and assign them.

CHE-1205 · CHEMISTRY-II · TOPIC 1 · LEVEL 1 TERM 2

1. The idea

Isomers are compounds with the same molecular formula but different structures or spatial arrangements — and therefore different properties. The formula C_2H_6O, for instance, is both ethanol (a liquid boiling at 78 °C, miscible with water, reactive with sodium) and dimethyl ether (a gas boiling at −24 °C, inert to sodium). Isomerism is why a molecular formula alone never identifies an organic compound, and why isomer counting is a standard examination sport. The master division:

BRANCH	DIFFERENCE LIES IN	INTERCONVERSION
Structural (constitutional)	Connectivity — which atom is bonded to which	Requires breaking/forming bonds
Stereoisomerism	Arrangement in space, same connectivity	Geometrical: needs bond breaking; Optical: impossible without breaking bonds

2. Structural isomerism

- **Chain isomerism** — different carbon skeletons. Butane C_4H_{10}: n-butane and 2-methylpropane (isobutane). Pentane C_5H_{12} has three: n-pentane, 2-methylbutane, 2,2-dimethylpropane. Hexane C_6H_{14} has five — the count grows combinatorially.
- **Position isomerism** — the functional group or multiple bond sits at a different position: but-1-ene vs but-2-ene; propan-1-ol vs propan-2-ol.
- **Functional-group isomerism** — different functional groups entirely: ethanol vs dimethyl ether (alcohol/ether); propanal vs propanone (aldehyde/ketone, C_3H_6O); acetic acid vs methyl formate (acid/ester, $C_2H_4O_2$).
- **Metamerism** — different alkyl groups around the same polyvalent functional group: diethyl ether vs methyl propyl ether (both $C_4H_{10}O$ ethers).

- **Tautomerism** — a special, dynamic case: isomers in rapid equilibrium by proton migration, classically keto ⇌ enol (ethyl acetoacetate is ≈ 93% keto, 7% enol at room temperature). Tautomers coexist in one flask; most other isomers must be separated deliberately.

3. Geometrical isomerism

Restricted rotation about a C=C bond (or in a ring) freezes two arrangements: **cis** (reference groups on the same side) and **trans** (opposite sides). But-2-ene: cis boils at 3.7 °C, trans at 0.9 °C; cis has a net dipole (≈ 0.33 D), trans cancels to zero — measurable differences from arrangement alone. When all four substituents differ, cis/trans runs out of language and the **E/Z system** takes over: rank the two groups at each end by Cahn–Ingold–Prelog priority (higher atomic number first); the two winners on the same side = *Z* (zusammen), opposite = *E* (entgegen). Geometrical isomers are **diastereomers** (see below) — different substances with different physical properties, separable by ordinary means.

4. Optical isomerism and chirality

A molecule is **chiral** if it cannot be superposed on its mirror image — the usual cause is a carbon bearing four different groups (an asymmetric or stereogenic centre, marked C*). The two mirror forms are **enantiomers**: identical in melting point, boiling point, solubility and spectra, differing only in the direction they rotate plane-polarised light (+/−, dextro/laevo) and in their behaviour toward other chiral things — enzymes, receptors, other enantiomers. Lactic acid, CH_3–C*H(OH)–COOH, is the textbook case; one enantiomer accumulates in sore muscles, the other appears in sour milk.

- **Racemic mixture:** equal amounts of both enantiomers — optically inactive by external compensation.
- **Diastereomers:** stereoisomers that are *not* mirror images (e.g. cis vs trans; or two-centre molecules with one centre flipped). Different physical properties, separable normally.
- **Meso compounds:** molecules with stereocentres that are achiral overall because of an internal mirror plane — tartaric acid's meso form is inactive by *internal* compensation. Hence the counting rule: maximum stereoisomers = 2^n for n stereocentres, but symmetry (meso forms) reduces the real count. Tartaric acid (n = 2): not 4 but **3** — (+), (−) and meso.
- **R/S assignment:** rank the four groups at C* by CIP priority, orient the lowest away, and read $1 \to 2 \to 3$: clockwise = R, anticlockwise = S. R/S is a label of arrangement; it does not by itself tell the sign of rotation.

5. Key points

- Isomers: same formula, different structure or arrangement — hence different properties.

- Structural: chain, position, functional group, metamerism, tautomerism (dynamic equilibrium).
- Geometrical: cis/trans for alike pairs, E/Z (CIP priorities) in general; geometrical isomers are diastereomers.
- Optical: chirality from non-superposable mirror images; enantiomers differ only in rotation sign and chiral environments.
- Counting: at most 2^n stereoisomers for n stereocentres; meso symmetry reduces the count; racemate = external, meso = internal compensation.

6. Practice

Answer in your own words; brief model answers follow.

1. Write the three chain isomers of C_5H_{12} and name each.
2. Ethanol and dimethyl ether share the formula C_2H_6O, yet one is a liquid reacting with sodium and the other an inert gas. Which type of isomerism is this, and what structural feature explains the reactivity difference?
3. Explain why but-1-ene shows no geometrical isomerism while but-2-ene does, and why the cis form of but-2-ene has a dipole moment but the trans form does not.
4. Tartaric acid has two asymmetric carbons, yet only three stereoisomers exist. Account for the missing fourth.
5. Identify the asymmetric carbon in 2-butanol and state the maximum number of its stereoisomers.
6. A solution contains equal amounts of (+)- and (−)-lactic acid. Is it optically active? Name the term for such a mixture and the kind of compensation involved.

Answer key

1. n-pentane $CH_3CH_2CH_2CH_2CH_3$; 2-methylbutane $(CH_3)_2CHCH_2CH_3$; 2,2-dimethylpropane $C(CH_3)_4$. Increasing branching lowers the boiling point across the family (36 → 28 → 9.5 °C) — a standard consequence.
2. Functional-group isomerism. Ethanol carries an –OH group: its O–H proton is acidic enough to react with sodium (hydrogen evolved) and hydrogen bonding makes it a high-boiling liquid. The ether's oxygens carry only C–O bonds — no O–H, no sodium reaction, weak intermolecular forces, a gas.
3. Geometrical isomerism needs each double-bond carbon to carry two different groups. In but-1-ene the terminal carbon carries two hydrogens — swapping them changes nothing. In but-2-ene each carbon carries H and CH_3, so cis and trans exist. In cis-but-2-ene the two C–CH_3 bond dipoles point to the same side and add (≈ 0.33 D); in trans they oppose and cancel.
4. The (2R,3S) form has an internal mirror plane through the middle of the molecule: it is superposable on its mirror image — a meso form, achiral despite two stereocentres. So the set is (2R,3R), (2S,3S)

[an enantiomeric pair] and meso (2R,3S): three, not $2^2 = 4$.

5. C-2 bears –OH, –H, $–CH_3$ and $–C_2H_5$: four different groups → asymmetric. n = 1 → at most $2^1 = 2$ stereoisomers, an enantiomeric pair.
6. No — the rotations cancel exactly. It is a racemic mixture (racemate); the compensation is external (between different molecules), unlike the internal compensation of a meso compound.
